Two-Block Circuit Formation under Langevin Annealing: Stationary Theory versus Training

Result in one sentence. Annealed Langevin noise at $T_0 = 10^{-7}$ does not change which circuit training selects (joint training still settles in the direct phase H_D, parent-warm in the compositional phase H_C), and picking the lower-objective run per cell gives H_C in only 29 of the theory's 47 cells.

Experiment. Same objective, network (bias-free two-block ReLU, $N = 1024$, float64, exact four-bit cube, $\gamma_2 = \gamma_4 = 2$) and 8×9 decay grid as the deterministic report ($\lambda_E = \lambda_2 = \lambda_p = \lambda_4 = \lambda_0$, $\lambda_1 = (\lambda_1/\lambda_2)\lambda_0$), seed 0, both protocols. The final phase is plain SGD (momentum 0) with Langevin noise added after every update, $\Delta\theta = \sqrt{2\eta_g T(t)}\,\xi$, $\xi \sim \mathcal{N}(0, I)$, using each optimizer group's rate ($\eta_{E,V} = 10^{-2}$, readout $\eta_W = 10^{-2}/N$), so the stationary density is $\propto e^{-J/T}$. $T = T_0$ for 5k steps, decays exponentially to $10^{-3}T_0$ over 25k steps, and is 0 for the last 20k; every cell runs exactly 50k steps. Parent-warm cells start from the deterministic sweep's own seed-0 warm starts (noise-free pretraining, momentum 0.9).

New measurements. A block-2 neuron is a 2+2→4 unit when $\geq 90\%$ of $\mathrm{Var}\, z_n$ lies on the parent modes x_0x_1, x_2x_3 with balance $\min / \max \geq 0.8$, and it produces the quad mode with $|a_n| \geq 10^{-6}$ (units below this on every target mode are dead). With a_n the neuron's $d4$ amplitude and $w_{4,n}$ its readout weight,

$$\varphi_{\mathrm{out}} = \frac{\sum_{2+2\to4} |w_{4,n}a_n|}{\sum_{\mathrm{block\,2}} |w_{4,n}a_n|}, \qquad \varphi_{\mathrm{sq}} = \frac{\sum_{2+2\to4} a_n^2}{\sum_{\mathrm{block\,2}} a_n^2}, \qquad f_C^{\mathrm{sq}} = \frac{\sum_{\mathrm{block\,2}} a_n^2\,\pi_n}{\sum_{\mathrm{block\,1}} a_n^2 + \sum_{\mathrm{block\,2}} a_n^2},$$

where π_n is the parent share of $\mathrm{Var}\, z_n$ (the weight f_C uses). Theory: $\varphi = 1$ in H_C, 0 otherwise.

	Parent-warm		Joint		Lower objective	
Seed 0, 8×9 grid	determ.	annealed	determ.	annealed	determ.	annealed
Phase agreement with theory	87.5%	79.2%	37.5%	36.1%	75.0%	72.2%
Gate $H_C \Leftrightarrow R > 1/\sqrt{8}$ (active cells)	89.5%	96.2%	49.1%	34.0%	94.7%	83.0%
Mean $\lvert f_C - f_C^{\mathrm{theory}}\rvert$	0.147	0.162	0.340	0.350	0.199	0.186
Cells in H_C (theory: 47)	46	48	3	3	29	29
Cells won (parent-warm / joint)					30/42	33/39
Protocols select different phases	44/72 cells (deterministic) → 48/72 cells (annealed)					

Observations. (i) *Basin selection is unchanged.* Joint training keeps H_D in every cell where it had it except two; parent-warm changes 8 cells. Three move $H_D \to H_C$ inside the theory's H_D region ($\lambda_1/\lambda_2 = 20$ at $\lambda_0 \leq 3 \times 10^{-4}$, and $(10^{-2}, 5)$), which lowers phase agreement even though the measured-R gate now predicts the parent-warm phase in 96% of active cells. Five strong-decay cells ($\lambda_0 = 0.1$–0.3) end as *parents_only*: their $d4$ amplitude is still rising at 50k steps ($A_4 = 0.005$–0.22), whereas four of them reached $A_4 = 0.63$–0.78 with momentum 0.9 (the fifth was suppressed). Momentum-free SGD moves $\sim 10\times$ more slowly, so these cells are unfinished rather than noise-destroyed. (ii) *Where H_C forms, theory matches at moderate decay.* The largest parent-warm share, $f_C = 0.738$ at $(\lambda_0, \lambda_1/\lambda_2) = (10^{-2}, 1)$, sits next to the predicted 0.749. At $\lambda_0 \leq 3 \times 10^{-4}$, however, $f_C = 0.46$–0.49 regardless of the ratio, whereas the theory predicts 0.49–0.78 rising with the ratio. (iii) *Block 2 is compositional only once decay prunes it.* Inside the theory's H_C region the parent-warm φ_{out} grows with λ_0: 0.44, 0.47, 0.55, 0.72 at $\lambda_0 = 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}$, then 0.80–0.89 for $\lambda_0 \geq 10^{-2}$ (theory: 1). At weak decay half of block 2's $d4$ output comes from mixed or raw-bit neurons. In H_D cells $\varphi_{\mathrm{out}} \approx 0$, as predicted. (iv) *Output-weighted and block-intrinsic shares agree.* Across cells $\mathrm{corr}(\varphi_{\mathrm{out}}, \varphi_{\mathrm{sq}}) = 0.99$ and $\mathrm{corr}(f_C, f_C^{\mathrm{sq}}) = 0.99$; the readout-free values are higher at weak decay ($f_C^{\mathrm{sq}} \approx 0.60$, $\varphi_{\mathrm{sq}} \approx 0.53$), so the readout slightly favours the non-compositional routes there, and they converge for $\lambda_0 \geq 10^{-2}$ (Fig. 2). (v) *The lower-objective run is often H_D where theory predicts H_C* (Fig. 1, bottom row). Of the 18 such cells, 7 at $\lambda_0 \leq 10^{-3}$ are near-ties (joint's H_D is lower by 0.02–0.3% of J); 5 at $\lambda_0 = 10^{-2}$–0.1, $\lambda_1/\lambda_2 = 0.5$–$2$ favour H_D by 0.8–8%, a genuine discrepancy at least with the H_C minima that training reaches; and 6 strong-decay cells are unfinished in both runs. The selected H_C region is the low-ratio band plus a wedge up to $\lambda_1/\lambda_2 \approx 1$–$2$ at $\lambda_0 = 10^{-3}$–3×10^{-2}; the deterministic runs give the same count (29). (vi) *Losses do not discriminate the phases.* Final MSE is set mainly by λ_0 (below 10^{-5} at $\lambda_0 = 10^{-4}$, 10^{-3}–10^{-1} at $\lambda_0 = 0.1$, and 1 where the target is lost) and is nearly identical for the two protocols (Fig. 3): H_C and H_D fit equally well, so the circuit is selected by the regularizer and the dynamics, not by the data fit.

Limits. One seed and a single temperature. $T_0 = 10^{-7}$ is 10^3–10^6 times smaller than the objective differences between the two protocols' minima in the deterministic sweep (10^{-4}–10^{-1}), and no joint cell left H_D, so at this temperature the noise only explores within a basin. The fixed 50k plain-SGD steps also leave the strongest-decay cells unfinished. The next tests are a T_0 sweep (or a longer hold) and enough noise-free steps for strong-decay cells to converge.

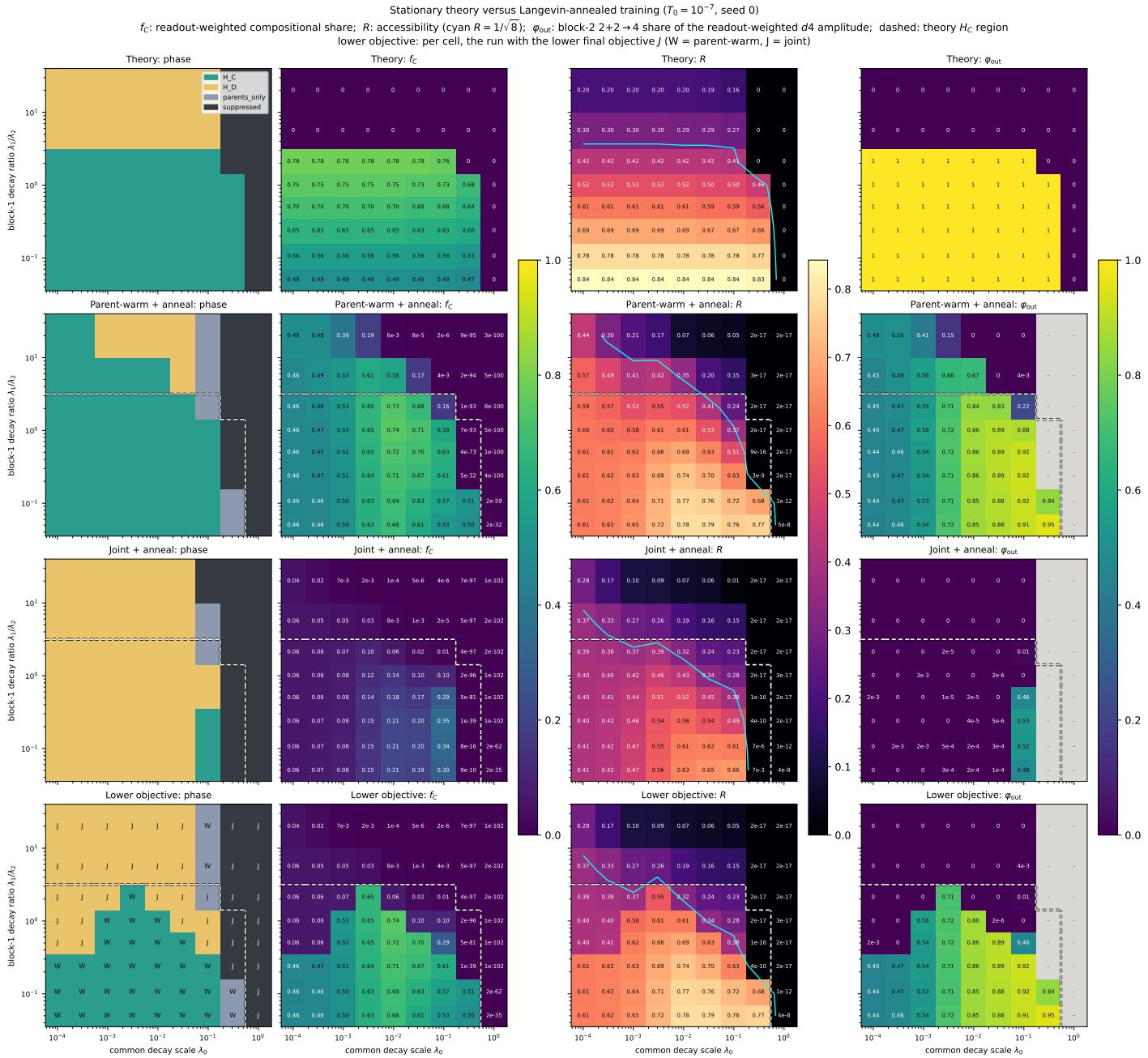


Figure 1: **Theory, parent-warm + anneal, joint + anneal, and the lower-objective run per cell (rows, top to bottom).** In the bottom row each cell shows whichever of the two annealed runs ended with the lower final objective J (W = parent-warm, J = joint). Columns: phase, readout-weighted compositional share f_C, accessibility R (cyan: $R = 1/\sqrt{8}$), and the block-2 2+2→4 share φ_{out} of the readout-weighted $d4$ amplitude. Dashed: theory H_C region. Grey: no target mode produced (collapsed network).

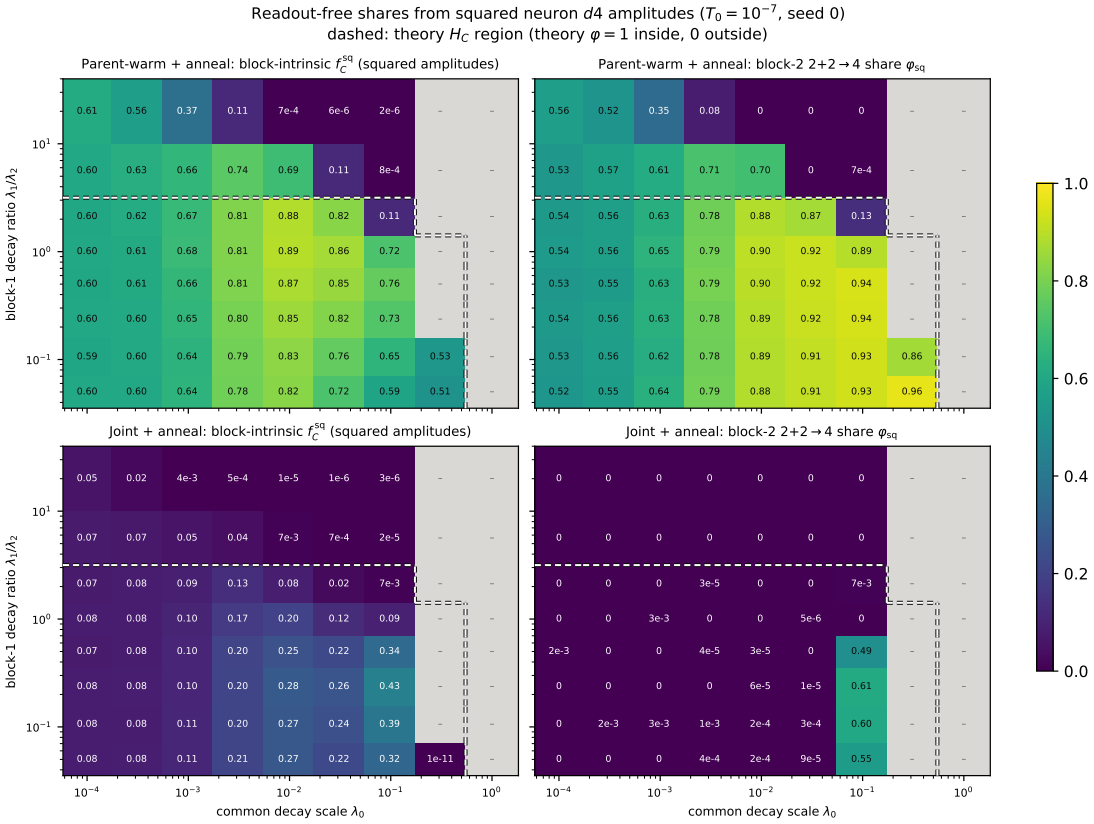


Figure 2: **Readout-free shares** from squared neuron $d4$ amplitudes: f_C^{sq} (left) and φ_{sq} (right); compare with columns 2 and 4 of Fig. 1.

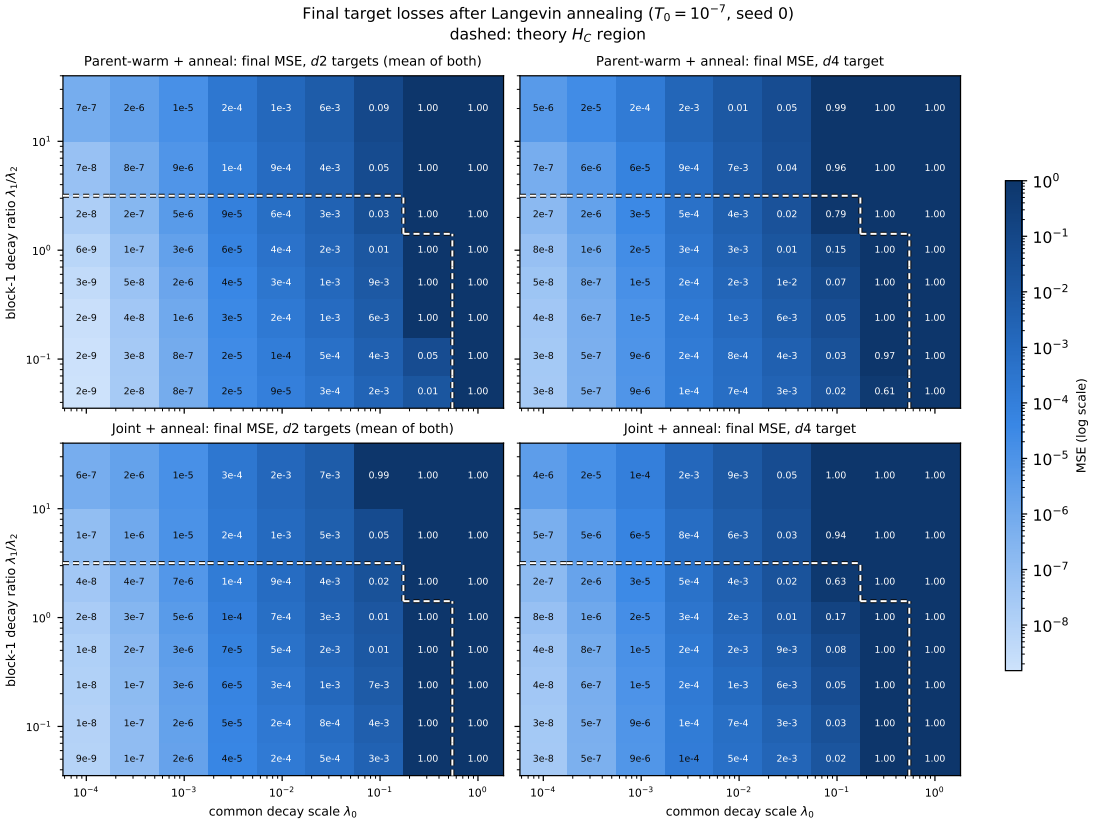


Figure 3: **Final target losses:** MSE of the two $d2$ targets (mean) and of the $d4$ target, log scale.